

# Kartik Agrawal

I work the full path from raw sensor measurements to learned action policies: state estimation at the bottom, systems integration in the middle, policy training at the top. Three years of production software engineering underneath it.

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## Estimation

*Where measurements become a state you can act on.*

🕒 **In progress**   Jul 2026 — Jan 2027

### Decentralized cooperative localization for UAV teams

Master's thesis · Institute of Automatic Control (IRT), RWTH Aachen

A framework for drones to improve their own state estimates from their neighbours. Each agent localizes from onboard GNSS, IMU and magnetometer; pairwise UWB ranging couples agents, reconciled with MESA — manifold edge-based separable ADMM.

GTSAM   MESA / C-ADMM   UWB   GNSS   ROS2   ArduPilot

#### Framework and open questions

- Single-agent GNSS/IMU/magnetometer factor graph estimator in GTSAM, validated at 0.3 m position RMSE against simulated ground truth.
- Real-time GNSS simulator with live ephemeris data and configurable error models, supporting loosely and tightly coupled architectures for comparative navigation studies.
- MESA's edge variables were designed for co-observed landmarks. Here agents share no landmarks — the coupled variable is the neighbour's pose, so the shared-variable set is defined by the ranging topology rather than by overlapping observation.
- Consensus ADMM chosen for asynchrony: dropouts and message loss appear as iterations where affected agents simply do not communicate, which matches real radio behaviour better than a synchronous scheme.
- Deliberately modality-agnostic — UWB is the first inter-agent measurement; depth or LiDAR can be added as further edge modalities.
- Built an ArduPilot SITL/ROS2 simulation environment with teleoperation, and a ROS2–React interface for state visualization and diagnostics.
- UWB simulation node implemented; the MESA interface is the current work.
- Open: batch versus incremental optimization, and the bandwidth cost of edge-variable exchange at scale.

## Integration

*Where components become a system that flies.*

● **Working**   Feb 2026 — present

### UAV autonomy stack

ROS2, PX4 SITL, Gazebo, Nav2

A full-stack autonomous UAV system in ROS2/PX4, supporting manual gamepad control and perception-driven autonomous flight in offboard mode.

ROS2   PX4   Nav2   LiDAR   C++   Docker   GitHub Actions

Code [https://github.com/Bravo1311/ROS2\\_PX4\\_Drone\\_Autonomy\\_POC](https://github.com/Bravo1311/ROS2_PX4_Drone_Autonomy_POC)   Demo <https://www.youtube.com/@Drononomy>

#### What it does

- ArUco-based visual localization with PD closed-loop velocity control for waypoint navigation and precision landing.
- LiDAR collision avoidance filtering commanded velocity in obstacle-facing directions.
- Nav2 adapted to UAV dynamics, enabling autonomous path planning and goal navigation on a drone platform.
- C++ marker-map annotator producing semantic annotations that feed VLM-based reasoning into the estimation and control layers.

- Every change is built automatically in a clean Docker container (GitHub Actions), so the project builds the same way on any machine. Changes go through pull requests.

## Policy

*Where the state estimate becomes an action.*

● Working Aug 2026 — present

### Flow-matching UAV autonomy

PyTorch, MuJoCo, Gazebo/PX4

A learned policy that flies the aircraft directly. A Diffusion Transformer-style conditional flow-matching model, trained by imitation from a classical PD controller and validated in both MuJoCo and Gazebo/PX4.

PyTorch Flow matching Transformers MuJoCo Imitation learning

Code [https://github.com/Bravo1311/Flow\\_Matching\\_UAVs](https://github.com/Bravo1311/Flow_Matching_UAVs) Demo [https://www.youtube.com/watch?v=w6VsROLk\\_S0](https://www.youtube.com/watch?v=w6VsROLk_S0)

#### *Architecture and the Phase 2 plan*

- Full pipeline: simplified MuJoCo dynamics, synthetic data generation, sliding-window dataset construction, AdaLN-conditioned transformer, closed-loop Euler-sampling inference.
- Trained against a classical controller rather than human demonstrations, so the target behaviour is exactly specified and failures are attributable to the policy rather than to the data.
- Phase 1 — privileged pose-based precision landing — is complete and public.
- Phase 2 in progress: vision-conditioned target following with a Siamese tracking head, cross-attention between template and search frame tokens, AdaLN conditioning from the template embedding, and localization output feeding the history encoder.
- Deliberate choice to train the tracking head separately before integrating with the DiT, so the two failure modes stay separable.

*Where this goes next*

### Closing the loop between the tracks

The three tracks are one system. The intended convergence is vision-based relative pose from the tracking policy entering the factor graph as an aiding factor, tested under simulated GNSS spoofing — learned perception feeding classical estimation rather than replacing it. Designed, not yet built.

## Experience

Sep 2024 — Jan 2026

### Software Engineering Intern

T-Systems International, Aachen

Built a Timefold AI optimization engine for Rego policy-driven automated scheduling, and the streaming infrastructure behind it.

Go Timefold Redpanda PostgreSQL Grafana

#### *Scope of the work*

- Designed and implemented the scheduling optimizer with tuned hard and soft constraints.
- Developed tooling that reduced solver runtime and constraint violations, plus a web interface for results visualization.
- Containerized microservices for real-time data streaming using Redpanda, custom Go plugins, PostgreSQL persistence and AMQP for reliable event processing.
- Instrumented the system for observability with Grafana and Tempo.

Jul 2021 — Aug 2023

### Full-Stack Software Engineer

LTIMindtree, Bangalore

Production microservices and full-stack applications for client-facing projects.

Spring Boot MERN SQL Graph databases

#### *Scope of the work*

- Production microservices and full-stack applications using Spring Boot and the MERN stack.
- Took multiple proofs of concept from problem definition through demonstration to client approval and deployment.
- Worked with relational and graph databases on automation-focused solutions.

Feb 2021 — Jun 2021

### Area Management Intern, Fulfillment Center

Amazon India (ASSPL), Mumbai

Traced defect sources in the inbound shipment phase and automated pallet sorting.

VBA

Process analysis

### Scope of the work

- Identified gaps and defect sources in the inbound phase through area surveying and analysis of process and defect metrics.
- Deployed VBA macros automating shipment-wise pallet sorting and identification via coded pallet labels.

## Education

Oct 2023 — present

### M.Sc. Robotic Systems Engineering

RWTH Aachen University

Aug 2017 — Jul 2021

### B.Eng. Mechanical Engineering

BITS Pilani

**Relevant coursework — RWTH Aachen.** Advanced Machine Learning, Computer Vision, Simulation of Robotic Systems, Sensors and Environment, Reinforcement Learning, Mechatronics, Mobile Robotics, Industrial Logistics.

## Skills

### Estimation

Factor graph optimization, GTSAM,  
GNSS (GPS, Galileo), Inertial navigation,  
UWB ranging, Multi-sensor fusion,  
Consensus ADMM

### Robotics & simulation

ROS2, PX4, ArduPilot, Gazebo, MuJoCo,  
Nav2, MoveIt, Isaac Sim

### Machine learning

PyTorch, Flow matching, Transformers,  
Imitation learning, VLMs

### Programming Languages

Python, C++, Java, JavaScript (MERN), SQL

### Infrastructure

Git, Docker, GitHub Actions, Linux,  
Redpanda, PostgreSQL, Grafana

### Languages

English (fluent), German (A2), Hindi (native)